Integrated Project Report: Data Science & Machine Learning

Comprehensive Analysis of Four Distinct Projects

Synthesis of Five Uploaded Project Notebooks

October 18, 2025

Advanced Data Analytics

Integrated Project Portfolio Overview

Four Distinct Project Domains

- **Project 1 (HR):** Employee Attrition Prediction
- **Project 2 (Security):** CloudWatch Web Attack Detection
- **Project 3 (Safety/Operations):** Incident Cause Analysis
- **Project 4 (Finance):** Stock Market Trend Analysis

Table 1: Core Techniques per Project

Project	Core Business Focus	Key ML/Data Technique
P1: Attrition	Retention/HR Strategy	Classification, EDA
P2: Web Attack	Cybersecurity/Monitoring	Anomaly Detection, Deep Learning (C
P3: Incident Cause	Root Cause Analysis/Safety	Descriptive Analysis, Visualization

P1: HR Attrition Prediction

P1: Employee Attrition – Data & Initial Findings

- **Objective:** Predict employee turnover to inform targeted retention strategies.
- **Data Structure:** 1,470 records, 35 features. Confirmed clean: No missing values or duplicates (`PROJECT_2.ipynb`).

Critical Finding: Class Imbalance

The overall Attrition Rate is **16.12%**.

Demographic Insights (`PROJECT_3.ipynb`)

- **Age:** Analysis shows a distinct density/concentration of attrition at certain ages (KDE plot).
- **Gender:** Attrition Rate was explicitly calculated for Gender (Bar plot), indicating a potential difference between groups.

P1: Modeling Pipeline and Recommendations

Preprocessing Steps

- **Scaling:** Numerical features (e.g., Monthly Income) must be scaled.
- **Encoding:** Categorical features converted (e.g., One-Hot Encoding).
- **Statistics:** Mean age $\approx$ 36.9 years, Average Working Years $\approx$ 11.3 years.

Modeling & Evaluation

- **Model Types:** Random Forest Classifier (Ensemble) and Deep Learning (Sequential Models) are utilized.
- **Evaluation:** Focus on metrics robust to imbalance (e.g., F1-Score, AUC) for accurate risk assessment.

P2: Web Attack Detection

P2: CloudWatch Anomaly Detection

- **Objective:** Identify suspicious and anomalous web traffic patterns in CloudWatch logs.
- **Key Preprocessing:** Successful conversion of time features and standardization of IP country codes.

Advanced Feature Engineering

1. **Time Feature:** Calculated `duration_seconds` to measure traffic session length.
2. **Scaling:** Applied **StandardScaler** to numerical features (`bytes_in`, `bytes_out`).
3. **Encoding:** Used **OneHotEncoder** on categorical variables for model compatibility.

P2: Modeling – Hybrid Anomaly & Classification

Unsupervised Anomaly Detection

Isolation Forest (IF)

- Highly effective algorithm for flagging outliers in large, high-dimensional data, ideal for detecting zero-day or novel attacks.

Supervised Deep Learning

1D Convolutional Neural Network (CNN)

- Architecture includes `Conv1D`, `Flatten`, and `Dense` layers.
- Used alongside Random Forest for robust, binary or multi-class attack classification.

P3: Incident Cause Analysis

P3: Analyzing Incident Outcomes and Root Causes

- **Objective:** Quantify and visualize the relationship between the **Cause category** and the **Outcome of Incident** to inform safety/operational improvements.
- **Methodology:** Primarily based on descriptive statistics and visualization (Grouping and plotting).

Key Analytical Steps

1. Data loaded and prepared for grouping (assuming cleaning steps like null-handling completed).
2. Incidents are grouped by `Cause category` and `Outcome of Incident`.
3. Total `Count` of incidents summed per grouping.

P3: Visualizing Cross-Analysis

- **Chart Type:** Stacked Bar Chart was generated using `matplotlib` and `pandas.unstack()`.
- **Interpretation Focus:** The chart visualizes the distribution of different outcomes across the various cause categories.

Actionable Insight

By analyzing the tallest bars (highest incident counts) and the composition of the stacks (outcome distribution), we can pinpoint:

- The most frequent root causes.
- Which causes lead to the most severe/specific outcomes.

P4: Stock Market Trend Analysis

P4: Financial Trend Forecasting

- **Objective:** Analyze stock data (`stocks (1).csv`) using regression to build a hypothetical trading strategy.
- **Data Transformation:** The `Date` column was converted and set as the index, establishing the time-series structure.

Volatility Feature Creation

A key financial feature was engineered:

- **Daily Range:** Calculated as $\texttt{High} - \texttt{Low}$.
- **Purpose:** Quantifies intraday price volatility, essential for trading model features.

Visualization: Confirmed trends for Closing Price, Trading Volume, and Daily Range over the observation period.

P4: Strategy Backtesting and Results

Modeling

- **Model:** Linear Regression was used for initial price forecasting.
- **Metrics:** Evaluated using Mean Squared Error (MSE) and the R^2 Score.

Performance Metrics

Strategy Return vs. Market Benchmark

- Total Strategy Return: **-0.45**
- Total Market Return: **-0.31**

Conclusion: The strategy significantly underperformed the benchmark, indicating the need for improved model inputs or a more complex forecasting model.

Final Conclusion

Final Synthesis and Next Steps

Table 2: Summary of Deliverables

Project	**Key Insight/Model**	**Recommended Next Step**
P1: HR Attrition	16.12% Attrition Rate	Implement predictive model for targeted intervention.
P2: Web Attack	IF and CNN Pipeline Established	Integrate real-time log ingestion for live monitoring.
P3: Incident Cause	Outcome vs. Cause Category Analysis	Deep dive into high-frequency/high-impact cause categories.
P4: Stock Trend	Strategy Return (~**0.45**)	[illegible]